# CREWMATIC

## Property-Level Data Model

Product Specification v1.0

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## Purpose

The property-level data model is the foundational architecture that enables Crewmatic's core value proposition: a single property can have multiple jobs across its lifecycle, and critical spatial/structural data persists at the property level rather than being siloed per job.

Why this matters:

A flooded basement in January becomes a reconstruction project in March. The sketch drawn during mitigation is the same floor plan needed for reconstruction estimating. Without a property-level model, contractors would redraw the same house twice — wasting time and introducing errors.

**Critical principle:** This cannot be retrofitted. The data architecture must be designed for property-first from day one.

## The Problem Without Property-Level Data

### Current Industry Pattern (Encircle, CompanyCam, etc.)

Each job is an isolated record:

• Job #1: 123 Main St — Water Damage — Sketch A, Photos A, Equipment A

• Job #2: 123 Main St — Reconstruction — Sketch B (redrawn from scratch), Photos B

Problems:

• Tech redraws the floor plan for reconstruction (30-60 minutes wasted)

• Sketch measurements don't match between jobs (mitigation said 120 SF, reconstruction says 140 SF)

• Customer sees two separate projects with no connection

• Historical context is lost (can't see what was done during mitigation when quoting reconstruction)

### Crewmatic Property-First Pattern

One property record with linked jobs:

• Property: 123 Main St
  - Sketch: Floor plan (drawn once, used forever)
  - Job #1: Water Damage (references property sketch)
  - Job #2: Reconstruction (references same property sketch, adds reconstruction-specific data)

Benefits:

• Sketch drawn once, used for all jobs at that property

• Mitigation moisture readings inform reconstruction scope

• Customer portal shows complete property history

• Estimating is faster (sketch data already exists)

# Core Entity Relationships

## The Three-Tier Hierarchy

```
CUSTOMER
  └── owns multiple PROPERTIES
      └── each property has multiple JOBS over time
          └── each job references shared property data (sketch, address)
```

Example:

Customer: John Smith

Property #1: 123 Main St, Warren MI

Sketch: Floor plan (Kitchen 12×10, Basement 20×30, etc.)

Job #1: Water Mitigation (4/10-4/15/2026) — Used sketch for equipment/moisture

Job #2: Reconstruction (4/20-5/30/2026) — Inherited same sketch

Property #2: 456 Oak Ave, Sterling Heights MI

Sketch: Floor plan (different house)

Job #1: Mold Remediation (3/1-3/10/2026)

# Four Core Principles

## 1. Sketch Belongs to Property, Not Job

**The Rule:** One sketch per property, shared across all jobs.

**Why:** The house doesn't change between mitigation and reconstruction. Room dimensions, wall locations, and floor plan geometry are property attributes, not job attributes.

How It Works:

When a property is first created (usually during the first job):

• Property record is created with address and customer

• Sketch record is created and linked to that property

• First job references the property (and therefore has access to the sketch)

When a second job is created at the same property:

• System detects existing property record (by address match)

• New job is linked to existing property

• Sketch is automatically available — no redrawing needed

What Lives Where:

Property Sketch (shared):

• Room geometry (polygon coordinates)
• Room dimensions (L×W, ceiling height)
• Room types (Kitchen, Bathroom, Bedroom)
• Calculated square footages (floor, ceiling, wall)
• Material flags (drywall, carpet, tile — structural defaults)

Job-Specific (not shared):

• Equipment pin locations (this job's air movers)
• Moisture pin locations (this job's drying cycle)
• Photo pins (this job's documentation)
• Affected area flags (mitigation scope for this episode)

User Experience:

Job #1 (Mitigation):

• Tech opens job, sketch tool launches
• Tech draws floor plan (Kitchen 12×10, Basement 20×30)
• Sketch auto-saves to property record
• Tech places equipment pins, moisture pins on sketch
• Job completes

Job #2 (Reconstruction, same property):

• Office opens new job at 123 Main St
• System says: "Sketch already exists for this property. Load it?"
• Sketch appears with all rooms, dimensions intact
• Equipment/moisture pins from Job #1 are hidden (job-specific data)
• Can add reconstruction annotations (new wall locations, finish notes)
• Original sketch dimensions remain unchanged

Sketch Versioning:

The sketch can evolve over time:

• Mitigation tech draws Kitchen as 12×10
• During reconstruction, discovers Kitchen is actually 12×12
• Updates sketch dimensions → creates Version 2
• Version history preserved (V1 from mitigation, V2 from reconstruction)
• Both jobs reference same sketch, but see appropriate version

## 2. Customer Can Own Multiple Properties

**The Rule:** One customer record, many properties.

**Why:** Property management companies, landlords, and repeat customers work on multiple addresses. Customer contact info should update once, apply everywhere.

How It Works:

When creating a job:

1. System asks: "Is this a new customer or existing?"
2. If existing → select from customer list
3. System asks: "Which property?" → shows all properties owned by that customer
4. If new property → create property record linked to that customer
5. Job is created linked to property (and therefore customer)

Customer Detail View:

Shows all properties owned by this customer:

Customer: ABC Property Management
Contact: Sarah Johnson
Phone: (586) 555-1234

Email: sarah@abcproperties.com

Properties (3):

📍 123 Main St, Warren MI
Jobs: Water Mitigation (4/10/26), Reconstruction (4/20/26)

📍 456 Oak Ave, Sterling Heights MI
Jobs: Mold Remediation (3/1/26)

📍 789 Elm St, Warren MI
Jobs: Fire Restoration (2/15/26)

Benefits:

- Update customer phone once → applies to all properties
- See complete customer history (all properties, all jobs)
- Easier repeat business tracking
- Customer portal login shows all their properties

## 3. Jobs Are Sequential Episodes at a Property

**The Rule:** Jobs have temporal relationships — mitigation happens before reconstruction.

**Why:** Context from prior jobs informs future work. Knowing what was done during mitigation helps scope reconstruction accurately.

Property Timeline View:

Property: 123 Main St, Warren MI

Owner: John Smith

Timeline:

```
┌─ Job #1: Water Mitigation ──────────────────┐
│ 4/10/2026 - 4/15/2026                       │
│ Status: Completed ✓                         │
│ Sketch: Created (Kitchen, Basement)         │
│ Equipment: 4 air movers, 3 dehus            │
│ Final Readings: All dry                     │
│ Certificate: Issued 4/16/26                 │
└─────────────────────────────────────────────┘
        ↓
┌─ Job #2: Reconstruction ────────────────────┐
│ 4/20/2026 - In Progress                     │
│ Status: Phase 3 of 8 (Rough-In)             │
│ Sketch: Inherited from Job #1               │
│ Scope: Kitchen rebuild + Basement           │
│ Estimate: $42,500                           │
└─────────────────────────────────────────────┘
```

"Convert to Reconstruction" Workflow:

When mitigation is complete:

1. Tech/owner clicks [Convert to Reconstruction] on completed mitigation job
2. System creates new job linked to same property
3. Job type = Reconstruction

4. Sketch automatically available (inherited from property)
5. Can reference mitigation data (moisture readings, affected areas) when scoping reconstruction
6. Mitigation job remains in history (completed/archived)

Benefits:

• No data re-entry (sketch already exists)
• Can review mitigation scope to inform reconstruction estimate
• Customer sees connected history (mitigation → reconstruction)
• Accurate before/after documentation (mitigation photos vs. reconstruction completion photos)

## 4. Property-Specific vs. Job-Specific Data

This is the critical design decision that makes everything work.

### PROPERTY-LEVEL DATA (persists across all jobs)

Address:
• Street, city, state, ZIP
• Never changes unless property record is corrected

Customer Relationship:
• Who owns/manages this property
• Can change over time (property sold to new owner)

Sketch (Floor Plan):
• Room geometry (polygon coordinates)
• Room names (Kitchen, Bedroom 2, Basement)
• Room types (kitchen, bathroom, bedroom — drives material assumptions)
• Dimensions (L×W, ceiling height)
• Calculated SF (floor, ceiling, wall perimeter)
• Material flags (structural defaults: drywall, carpet, etc.)
• Floor levels (basement, main, upper, attic)

Property Notes:
• Gate code: 1234
• Key location: Under mat
• Access notes: Call before arriving
• Permanent info that applies to all jobs

Historical Job List:
• Every job ever performed at this property
• Chronological timeline

### JOB-LEVEL DATA (specific to one episode)

Job Metadata:
• Job number (unique identifier)
• Job type (water mitigation, fire, mold, reconstruction)
• Status (lead, in progress, completed)
• Dates (created, scheduled, started, completed)
• Assigned tech

Insurance Info:

• Claim number
• Carrier name
• Adjuster contact
• (Specific to this loss episode)

Equipment Records:
• Which air movers/dehus were placed
• Serial numbers
• Placement/pull dates
• Room locations (references property sketch rooms)

Moisture Readings:
• Daily readings per location
• Material types tested
• Trend data (improving/plateau/rising)
• Room locations (references property sketch rooms)

Photos:
• Damage documentation
• Equipment placement
• Before/after
• Room assignments (references property sketch rooms)

Estimate:
• Line items
• Pricing
• Totals
• (Specific to this job's scope)

Certificate:
• Completion proof
• Final readings
• Equipment summary
• (Issued for this job)

Reconstruction-Specific (if job type = reconstruction):
• Phases (Demo, Rough-In, Drywall, etc.)
• Material selections (cabinets, flooring, paint colors)
• Change orders (scope additions during reconstruction)
• Punch list (final items to complete)
• Milestone payments (tied to phase completion)

# How Data Flows Between Layers

## Scenario 1: Creating First Job at a Property

**User Action:** Create new water mitigation job at 123 Main St

System Behavior:

1. Checks: Does property "123 Main St" already exist?
   - No → Create new property record
2. Checks: Does customer "John Smith" already exist?
   - No → Create new customer record

- Yes → Link property to existing customer
3. Links property to customer
4. Creates job record linked to property
5. Sketch tool opens → Tech draws floor plan
6. Sketch saves to property (not job)
7. Equipment/moisture/photos save to job (with references to sketch rooms)

Result:

• Customer record created
• Property record created with sketch
• Job #1 created, linked to property

## Scenario 2: Creating Second Job at Same Property

**User Action:** Create reconstruction job at 123 Main St (mitigation already complete)

System Behavior:

1. Checks: Does property "123 Main St" already exist?
   - Yes → Found existing property record
2. System shows: "This property has a sketch and 1 prior job. Load existing data?"
3. User confirms → System links new job to existing property
4. Sketch automatically available (inherited from property)
5. Prior job visible in timeline (for reference)

User Experience:

Opening reconstruction job:

```
┌──────────────────────────────────────────────────────────┐
│ 123 Main St - Reconstruction Job                         │
│ ──────────────────────────────────────────────────────── │
│                                                          │
│ ℹ️ This property has existing data:                      │
│                                                          │
│ ✓ Sketch from Water Mitigation (4/10/26)                 │
│ ✓ Prior Job: Water Mitigation (completed)                │
│                                                          │
│ [Load Sketch] [View Prior Job] [Start Fresh]             │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

Most users click [Load Sketch] → Floor plan appears instantly.

## Scenario 3: Updating Sketch During Reconstruction

**User Action:** During reconstruction, tech discovers Kitchen is 12×12, not 12×10 as drawn during mitigation

System Behavior:

1. User edits Kitchen room dimensions in sketch tool
2. System detects change to property sketch
3. Creates new sketch version (Version 2)
4. Change logged: "Kitchen dimensions updated: 12×10 → 12×12 (Brett, 4/22/26)"
5. Both jobs can reference appropriate version:

- Mitigation job (completed) → locked to Version 1
- Reconstruction job (active) → uses Version 2

Why versioning matters:

• Mitigation estimate was based on 120 SF (12×10) — that's historical fact
• Reconstruction estimate is based on 144 SF (12×12) — accurate current measurement
• Audit trail shows when/why sketch changed

### Scenario 4: Customer Portal Shows Property History

**User Action:** John Smith logs into customer portal

What He Sees:

Welcome, John Smith

Your Properties:

```
┌─ 123 Main St, Warren MI ──────────────────────────────┐
│                                                       │
│  Jobs:                                                │
│                                                       │
│  Water Mitigation (4/10-4/15/2026)                    │
│    Status: Completed ✓                                │
│    Certificate: [Download PDF]                        │
│    Photos: 24                                         │
│                                                       │
│  Reconstruction (4/20-Present)                        │
│    Status: In Progress (Phase 3 of 8)                 │
│    Next Milestone: Drywall (Est. 5/5/26)              │
│    Photos: 18                                         │
│                                                       │
│  [View Floor Plan] [View All Photos]                  │
│                                                       │
└───────────────────────────────────────────────────────┘
```

All data is organized by property, with jobs nested underneath. Customer sees complete history at a glance.

## Handling Partial Sketches & Sketch Extensions

### The Problem

Scenario:

• Job #1 (Mitigation): Basement flooded. Tech sketched only the basement (Kitchen upstairs unaffected, not drawn).
• Job #2 (Reconstruction): Homeowner wants to remodel the kitchen (which was never sketched).

**What should happen:** Tech needs to add the kitchen to the existing sketch rather than start over.

### Solution: Sketch is Additive, Not All-or-Nothing

#### Core Principle

**The sketch is a living document that grows over time as new areas of the property are documented.**

• Job #1 sketches the basement → Sketch has 1 floor level (basement)
• Job #2 needs the kitchen → Tech adds main floor level, draws kitchen
• Job #3 needs upstairs bedrooms → Tech adds upper floor level, draws bedrooms

**The property sketch becomes more complete with each job.**

## User Experience: Adding Rooms to Existing Sketch

### *When Opening Job #2 (Kitchen Remodel)*

User opens reconstruction job, clicks into Sketch tab:

```
┌──────────────────────────────────────────────────────────────┐
│ Sketch - 123 Main St                                         │
│ ──────────────────────────────────────────────────────────── │
│                                                              │
│ Existing Sketch Found (from Water Mitigation):               │
│ • Basement (3 rooms)                                         │
│                                                              │
│ This job needs: Kitchen (Main Floor)                         │
│                                                              │
│ ○ Add to existing sketch (recommended)                       │
│   Continue building on the existing floor plan               │
│                                                              │
│ ○ Start new sketch                                           │
│   Create separate floor plan (not recommended)               │
│                                                              │
│              [Cancel]  [Continue]                            │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

### *Option 1: Add to Existing Sketch (Recommended)*

User selects "Add to existing sketch" → Opens sketch tool:

What they see:

• Floor selector at top: Basement | Main Floor | Upper Floor | Attic
• Currently viewing: Basement (shows existing rooms from Job #1)
• Tap Main Floor → Switches to empty canvas
• Message: "Main floor not yet sketched. Add rooms to this level."
• User draws kitchen, dining room, living room as needed
• Saves → Sketch now has both basement (from Job #1) and main floor (from Job #2)

Sketch Version Created:

• Version 2: "Main floor added (Kitchen, Dining Room, Living Room) — Brett, 4/25/26"
• Job #1 (mitigation) still references Version 1 (basement only)
• Job #2 (reconstruction) references Version 2 (basement + main floor)

### *Option 2: Start New Sketch (Edge Case)*

User selects "Start new sketch" → Creates separate sketch for this job:

What happens:

• New sketch record created (linked to property, but distinct from Job #1 sketch)
• Job #2 uses new sketch

• Job #1 still references original sketch
• Property now has two sketches (unusual, but supported)

When this makes sense:

• Property is multi-unit (Duplex: Unit A basement sketched for Job #1, Unit B kitchen for Job #2)
• Property is commercial with separate buildings (Building 1 sketched for Job #1, Building 2 for Job #2)
• Prior sketch was inaccurate/unusable, starting fresh is easier

User Warning:

⚠️ Starting a new sketch means this job won't share
floor plan data with prior jobs. This is unusual.

Are you sure? Most contractors add to the existing
sketch instead.

[Go Back]  [Start New Sketch Anyway]

# Address Matching Logic

**Problem:** How does system know if "123 Main St" from a new job is the same property as existing "123 Main Street"?

**Solution:** Fuzzy address matching with manual override.

## Auto-Match Behavior

When creating a new job, system checks existing properties for matches:

Exact Match:

• Same street number AND street name AND city AND ZIP
• Example: "123 Main St" = "123 Main Street" (abbreviation normalized)
• System auto-suggests: "Use existing property at 123 Main St?"

Close Match:

• Same street number AND similar street name AND same ZIP
• Example: "123 Main St" vs. "123 Main Street Apt 2"
• System asks: "Is this the same property as 123 Main St? [Yes] [No, create new]"

No Match:

• Different address entirely
• System creates new property record

## Manual Override

User can always:

• Force creation of new property (even if address matches)
• Merge duplicate properties later (Settings > Properties > Merge Duplicates)

Why manual override matters:

• Duplexes: "123 Main St Unit A" and "123 Main St Unit B" are separate properties
• Multi-building complexes: Same address, different buildings
• Data entry errors: Tech typo'd address, needs correction

# Property Detail Page (New UI Screen)

This is a new screen that doesn't exist in current specs — needed to manage property-level data.

## Location

Dashboard → Properties tab (new nav item)

OR

Job Detail → [View Property] button

## What It Shows

```
┌──────────────────────────────────────────────────────────────────────────────────────┐
│ Property: 123 Main St, Warren MI 48089                                               │
│ Owner: John Smith • (586) 555-1234                                                   │
│ ──────────────────────────────────────────────────────────────────────────────────── │
│                                                                                      │
│ [Sketch] [Jobs] [Photos] [Notes]                                                     │
│ ═══════                                                                              │
│                                                                                      │
│ Floor Plan:                                                                          │
│  ┌──────────────────────────────────────────────┐                                   │
│  │ [Sketch thumbnail]                           │                                   │
│  │ Last updated: 4/22/2026                      │                                   │
│  │ Rooms: Kitchen, Basement, Bedroom            │                                   │
│  └──────────────────────────────────────────────┘                                   │
│ [Open Sketch Editor] [View Sketch History]                                           │
│                                                                                      │
│ Jobs at This Property:                                                               │
│ ──────────────────────────────────────────────────────────────────────────────────── │
│ Job #1: Water Mitigation                                                             │
│   4/10-4/15/2026 • Completed ✓                                                       │
│   [View Job Details]                                                                 │
│                                                                                      │
│ Job #2: Reconstruction                                                               │
│   4/20-Present • In Progress (Phase 3/8)                                             │
│   [View Job Details]                                                                 │
│                                                                                      │
│ [+ Create New Job at This Property]                                                  │
│                                                                                      │
│ Property Notes:                                                                      │
│ ──────────────────────────────────────────────────────────────────────────────────── │
│ Gate code: 1234                                                                      │
│ Key under back door mat                                                              │
│ Dog in yard — call before entering                                                   │
│ [Edit Notes]                                                                         │
│                                                                                      │
└──────────────────────────────────────────────────────────────────────────────────────┘
```

### Tabs

Sketch Tab:

• Shows property sketch
• Can edit sketch (creates new version)
• View sketch version history

Jobs Tab:
• Chronological list of all jobs at property
• Click job → opens Job Detail page

Photos Tab:
• All photos from all jobs at this property
• Filterable by job
• Grouped by room (from sketch)

Notes Tab:
• Property-level notes (gate code, access info, etc.)
• Not tied to any specific job

## Success Criteria

Property-level data model is done when:

✅ Properties exist as separate entities from jobs
- Property has address, customer, sketch
- Jobs link to property (not duplicate address data)

✅ Sketch belongs to property, shared across jobs
- Sketch created once during first job
- Second job at same property auto-loads sketch
- Equipment/moisture/photos are job-specific, but reference sketch rooms

✅ Customer can own multiple properties
- Customer detail page shows all properties
- Properties list shows all jobs per property

✅ Address matching works
- System detects existing properties when creating new job
- Fuzzy matching handles "Main St" vs. "Main Street"
- Manual override available

✅ Property detail page exists
- Shows sketch, jobs, photos, notes
- Can edit property-level data
- Timeline shows all jobs chronologically

✅ "Convert to Reconstruction" workflow works
- Creates new job linked to same property
- Sketch automatically available
- Can reference mitigation data when scoping reconstruction

✅ Sketch versioning tracks changes
- Editing sketch creates new version
- Version history preserved
- Jobs reference appropriate version

✅ Sketch can be extended over time
- Job #1 sketches basement only
- Job #2 adds main floor to same sketch
- Property sketch grows more complete with each job

## What This Unlocks

Once property-level data model is in place:

Portals become possible:
• Customer portal can show all jobs at their property
• Adjuster portal can see complete property history
• Reconstruction client can review mitigation scope

Reconstruction becomes streamlined:
• Sketch already exists (from mitigation)
• Can reference affected areas from mitigation
• Estimating is faster (dimensions already measured)

Better customer experience:
• See connected history (not isolated jobs)
• One floor plan, consistent across all work
• Professional documentation

Competitive advantage:
• Encircle can't do this (job-centric architecture)
• Faster reconstruction quotes (sketch exists)
• More accurate estimates (no re-measuring)

— End of Specification —